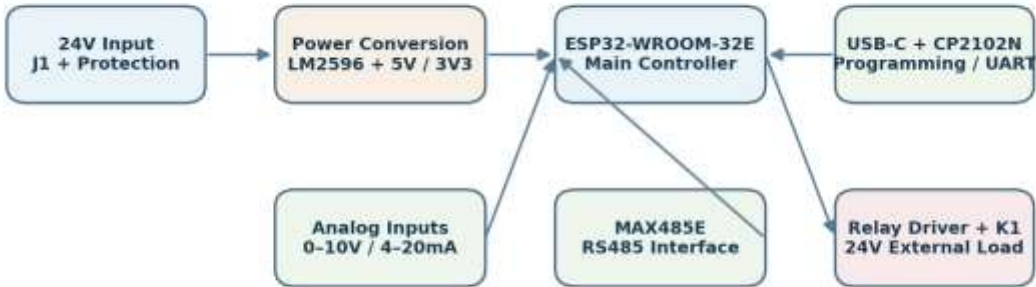


ESP32 Industrial I/O Controller

Complete PCB Design Process & Engineering Record

ESP32 Industrial I/O Controller — Functional Architecture



Document ID	EIC-4L-ENG-001
Documentation Revision	1.0
Design Revision	Rev A — Routed PCB
Toolchain	KiCad PCB + Schematic Editors
Board	4-layer FR-4, nominal 1.6 mm
Prepared for	Portfolio / GitHub / LinkedIn engineering record

Documentation note: This document records the design decisions and implementation state captured during the KiCad PCB development workflow. Values or performance limits not explicitly established in the project record are marked as design targets or items to verify before production.

Contents

1. Project Definition
2. Requirements
3. Architecture and Partitioning
4. Component Selection
5. PCB Stackup and Design Rules
6. Schematic Development
7. Footprint and Mechanical Decisions
8. PCB Placement Evolution
9. Routing Strategy and Execution
10. Verification and DRC
11. Final Gerber Inspection
12. Key Design Decisions and Change Log
13. Final Engineering Status
14. Portfolio Documentation Notes

1. Project Definition

Goal. Design a compact industrial-style ESP32 controller PCB that combines 24V field power, protected 5V/3.3V rails, USB service access, RS485, analog process inputs, and a relay-controlled external load interface in one 4-layer board.

- Controller core: ESP32-WROOM-32E.
- Service interface: USB-C with CP2102N USB-UART bridge.
- Industrial communications: MAX485E RS485 transceiver.
- Analog interfaces: one 0–10V input and one 4–20mA input.
- Field power: nominal 24V input with protection and fused board power.
- Outputs: 24V relay interface; external load target up to 5A inductive through relay contacts.
- PCB approach: 4 layers, solid inner GND plane, edge-accessible field connectors.

2. Requirements

Category	Requirement / target	Implementation
Power	24V field input	J1 → F1 → Q3 protection → 24V_PROTECTED
Power	5V rail	LM2596S-5 buck stage
Power	3.3V rail	TLV76733 linear regulator
USB	USB service/programming	USB-C + CP2102N
Analog	0–10V sensor	Protected divider/filter to GPIO34
Analog	4–20mA sensor	150 Ω shunt + filter/protection to GPIO35
Comms	RS485	MAX485E + A/B terminal
Output	Relay output	JQC-3FF-024-1Z SPDT relay
Load	External motor/load target	Up to 5A inductive at the external relay-contact path; verify final thermal/current margin before production
PCB	4-layer, 1.6 mm nominal	F.Cu / In1.Cu / In2.Cu / B.Cu

3. Architecture and Partitioning

ESP32 Industrial I/O Controller — Functional Architecture

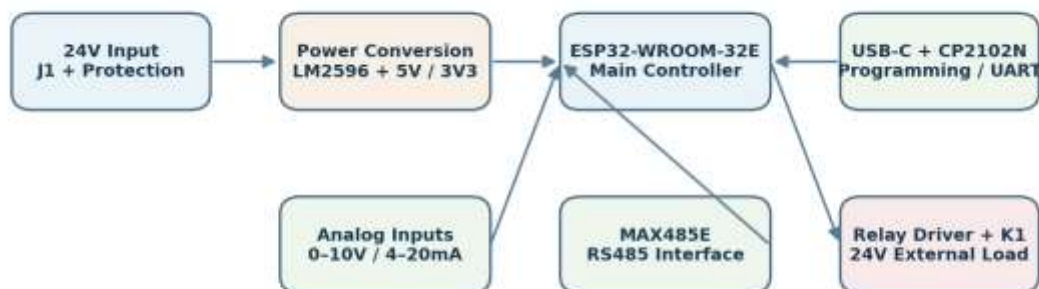


Figure 1. Functional architecture used to partition the PCB into power, digital, analog, communications, and relay zones.

- Power section kept in the lower portion of the board to isolate switching currents from analog and RF-sensitive circuitry.
- ESP32 placed near the upper center with its antenna facing the board edge and a dedicated keep-out region.
- Analog connectors placed at the left edge, with conditioning components immediately inboard.
- RS485 connector and MAX485E placed together at the right edge.
- Relay and external-load connector placed together in the lower-right zone.
- USB-C and CP2102N grouped at the upper-left service area.

4. Component Selection

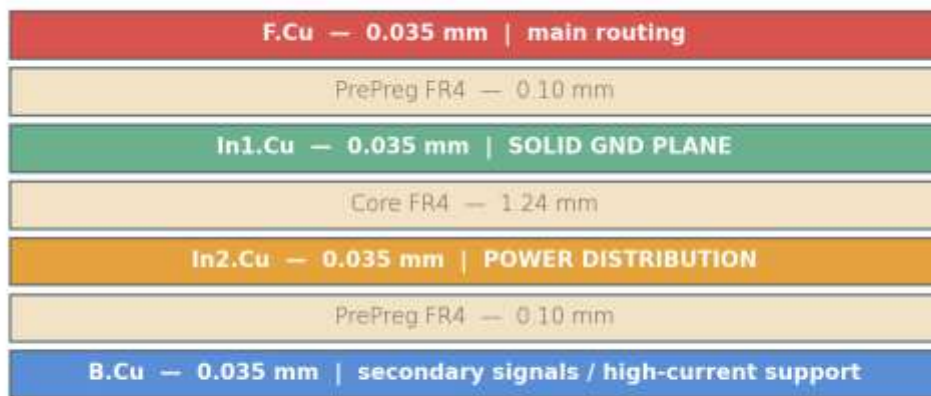
Core components and selected packages. The selection was made around the required electrical function, field accessibility, and KiCad footprint/3D-model availability.

Function	Part / value	PCB package / footprint decision
MCU	ESP32-WROOM-32E	RF_Module:ESP32-WROOM-32D style WROOM footprint used in the project
USB-UART	CP2102N	QFN-24 footprint with exposed GND pad
RS485	MAX485E	SOIC-8 footprint
5V buck	LM2596S-5	Power IC with external diode / inductor / capacitors
3.3V LDO	TLV76733DRVx	SOT-23-6 family footprint used for the project symbol/PCB mapping
Relay	JQC-3FF-024-1Z	Relay_THT:Relay_SPDT_Hongfa_JQC-3FF_0XX-1Z
24V field terminals	Phoenix Contact MKDS family	2-position 5.08 mm Phoenix footprint
3-way relay terminal	Phoenix Contact MKDS family	3-position 5.08 mm Phoenix footprint

Function	Part / value	PCB package / footprint decision
Resistors	10k, 20k, 47.5k, 150 Ω etc.	0805 metric for SMD resistors
Caps	0.1 μ F to 22 μ F SMD	0603/0805/1206 selected by value/function
Bulk caps	680 μ F / 220 μ F	Through-hole radial footprints

5. PCB Stackup and Design Rules

4-Layer PCB Stackup (not to scale)



Nominal finished board thickness: 1.6 mm

Figure 2. Final 4-layer conceptual stackup and assigned layer functions.

Layer / rule	Final project setting	Purpose
F.Cu	Signal / component side	Main routing and local power
In1.Cu	Solid GND plane	Continuous low-impedance return plane
In2.Cu	Power + selected routes	24V, 5V, 3V3 distribution
B.Cu	Signals + high-current support	Secondary routing and additional copper
Minimum clearance	0.20 mm	Primary spacing constraint
Minimum track width	0.20 mm	Board-level minimum
Minimum connection width	0.20 mm	DRC connection rule
Minimum via diameter	0.60 mm	Standard via preset
Through-hole minimum	0.20 mm after DRC adjustment	Accommodates small footprint holes; verify fab capability

Grounding decision. The final plan uses a continuous In1.Cu GND plane rather than an arbitrary split/star architecture. Local return paths remain short and return-current-aware.

6. Schematic Development

1. Functional blocks were drawn first: MCU, power tree, USB-UART, RS485, analog inputs, relay driver, and field connectors.
2. Nets were named to make PCB routing explicit: +3V3, +5V, VBUS, SYS_5V, 24V_PROTECTED, ANALOG_0_10V, ANALOG_4_20mA, RS485_A, RS485_B and control nets.
3. ERC cleanup included annotation/grid correction; the project reached zero schematic errors in the development workflow.
4. Footprints were then assigned to all components, with special attention to manufacturer-specific connector geometry and 3D models.
5. The schematic was used as the source of connectivity while PCB Editor handled physical placement and routing.

Block	Key implementation details
ESP32	EN/reset network, BOOT switch, RESET switch, local decoupling
3.3V	TLV76733 with input/output capacitors
USB	USB-C → CP2102N; USB support passives grouped locally
24V protection	J1 → F1 → PMOS Q3 + protection parts → 24V_PROTECTED
Buck	LM2596S-5 + 680 μ F input + Schottky diode + 33 μ H + 220 μ F output
Analog 0–10V	Protection/clamp + divider + 100 nF filtering to GPIO34
Analog 4–20mA	5V clamp + 150 Ω shunt + 100 nF filtering to GPIO35
RS485	MAX485E + bias resistors + 120 Ω termination
Relay	24V coil + Q4 BSS138 low-side drive + D5 flyback + J3 SPDT output

7. Footprint and Mechanical Decisions

- Generic 2-pin screw terminal footprints were replaced with manufacturer-specific Phoenix Contact MKDS family footprints to obtain a realistic mechanical envelope and usable 3D model.
- The same approach was applied to the 3-pin relay output terminal.
- Connector bodies were kept on board edges so field wiring can be inserted from outside the PCB without mechanical obstruction.
- The ESP32 antenna keep-out was preserved during every placement iteration.
- A nominal 1.6 mm board thickness was retained as the manufacturing/mechanical baseline.

Important production note. The Phoenix footprint selection records the PCB geometry, but the actual purchased part must still match the selected MPN and pitch during BOM procurement.

8. PCB Placement Evolution

Area	Initial condition	Final decision / change
USB	CP2102N was initially too far from USB-C	Moved U2 close to J2 and kept USB passives local
ESP32/RF	MCU centered with large antenna region	Kept at top/center with antenna at edge and keep-out preserved

Area	Initial condition	Final decision / change
Analog	Conditioning existed but had extra spacing	Grouped immediately behind J5/J6
RS485	U3 and connector separated during early placement	Grouped U3 + J4 + termination/bias parts
Buck	Components appeared visually spread in 2D	3D inspection showed acceptable physical proximity; no unnecessary compression
3.3V	U4/C1/C2 were reviewed for proximity	C1/C2 retained close to regulator
24V input	J1, F1 and protection parts at lower-left	Kept as a compact power-entry block
Relay	Relay and output connector at lower-right	Kept K1 + Q4 + D5 + driver passives + J3 together
Board outline	Started near 100 × 70 mm	Reduced after placement to remove large unused perimeter area

- Placement was deliberately refined using both 2D PCB Editor and 3D Viewer rather than optimizing only for visual compactness.
- After the design was compact enough, the central open area was retained as routing and isolation space instead of forcing the outline smaller.

9. Routing Strategy and Execution

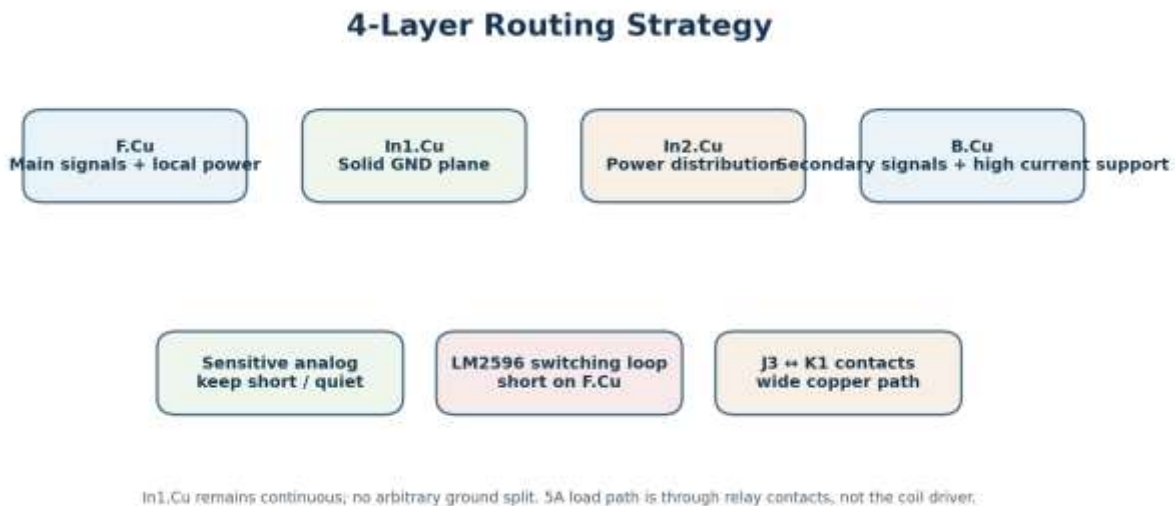


Figure 3. Layer-aware routing strategy used during execution.

1. LM2596 switching loop first, keeping the high-frequency loop compact on F.Cu and avoiding unnecessary vias.
2. 24V entry and protected power distribution next.
3. External motor/load path through the relay contacts using wide copper; the 5A load path is not routed through the relay coil driver.
4. 5V and 3.3V rail distribution, with local regulator capacitor connections kept short.
5. USB D+ / D- kept together and short between J2 and CP2102N.
6. RS485 A/B routed as a short coupled pair near U3/J4, with the 120 Ω termination kept close to the interface.
7. Analog channels routed short and away from switching/high-current regions.

8. Relay control wiring routed locally around Q4/D5/R11/R12.

9. Remaining ESP32 signals routed last, using B.Cu only when it improved routability.

Layer	Routing policy
F.Cu	Main signals, local power, LM2596 critical loop, analog, USB, RS485
In1.Cu	GND plane; no arbitrary split
In2.Cu	Main power distribution
B.Cu	Secondary signals and support for high-current copper

10. Verification and DRC

Check	Observed result / decision
PCB DRC violations	0 after the final check
Unconnected items	0
Schematic parity	An outstanding U4/footprint parity exception was ignored in the working release after PCB connectivity had already been verified; fix before a strict source-controlled release
Minimum drill	Rule adjusted to 0.20 mm to accommodate small footprint holes; confirm fab drill capability
Ground plane	In1.Cu verified as continuous GND plane in Gerber Viewer
Power layer	In2.Cu inspected for routed power copper
Gerbers	Opened in Gerber Viewer and reviewed layer-by-layer

Release discipline. A portfolio-ready record can show zero DRC and zero unconnected items while still disclosing source-parity exceptions. For production release, the schematic/PCB pair should be synchronized so the parity check also passes without exclusions.

11. Final Gerber Inspection

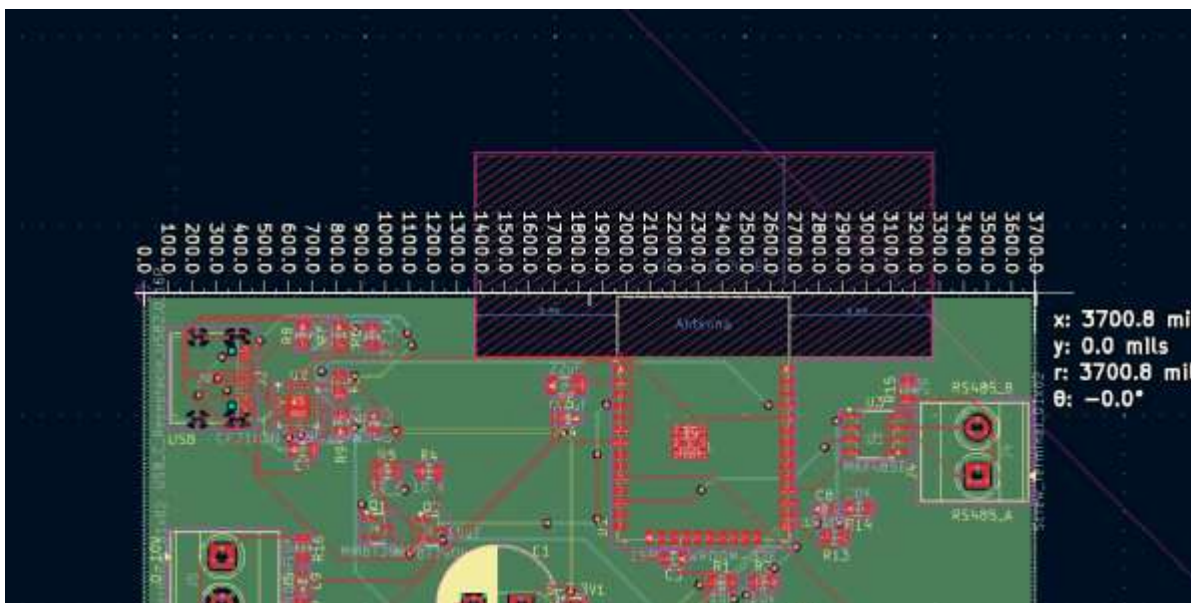


Figure 4. Gerber Viewer composite inspection — final board layers visible.

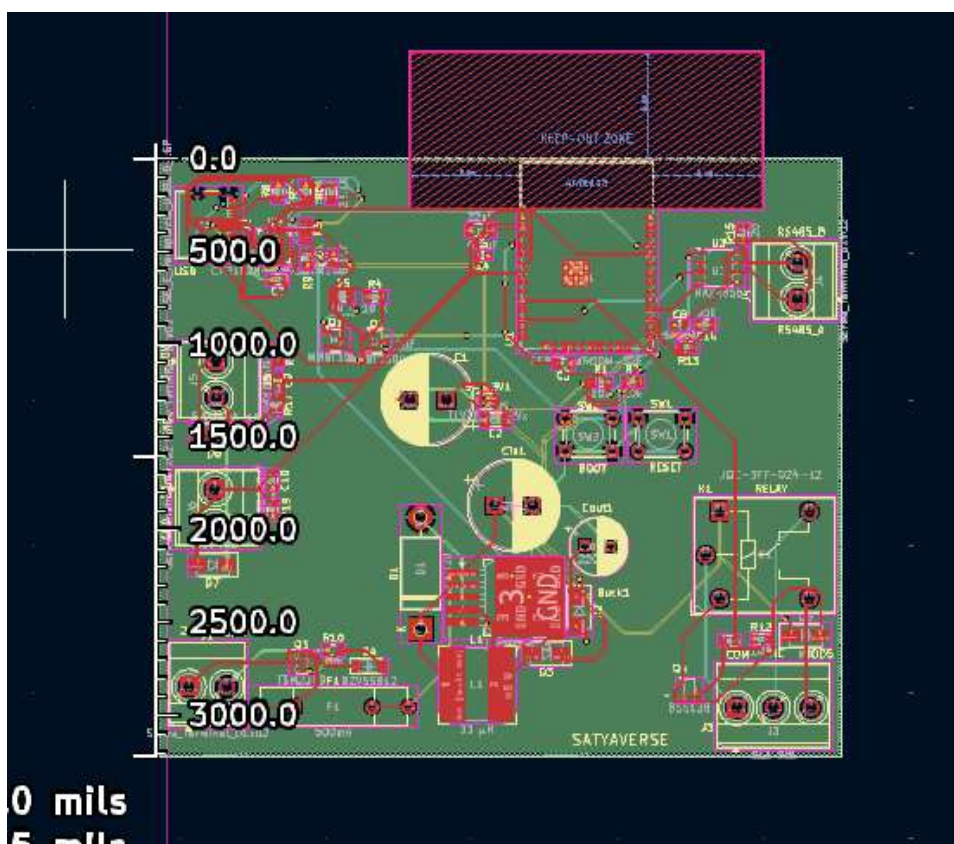


Figure 5. Gerber Viewer full-board inspection view used for final visual checking.

- Edge.Cuts observed as the board profile; the antenna keep-out remains a copper/rule area rather than an Edge.Cuts boundary.
- In1.Cu showed the intended continuous GND plane.
- In2.Cu showed the dedicated power routing layer.
- Front/back copper, masks, silkscreen, and drill outputs were inspected in Gerber Viewer.

12. Key Design Decisions and Change Log

Decision	Earlier approach	Final approach	Reason
Layer count	3-layer / 4-layer considered	4-layer	Dedicated GND plane + separate power layer
Connector footprint	Generic terminal block	Phoenix MKDS family	Mechanical realism + 3D model + field accessibility
Grounding	Split/star concepts considered during review	Continuous In1.Cu GND plane	Predictable return paths and easier controlled routing
USB placement	U2 had extra distance from J2	U2 close to J2	Shorter USB differential routing
Buck placement	2D view looked spread	Kept physically compact; no further forced compression	3D view confirmed adequate proximity
Board size	Initial ~100 × 70 mm	Reduced outline	Remove unnecessary perimeter only after placement stabilized
U4 capacitor placement	Reviewed repeatedly	C1/C2 kept at practical proximity	Avoid unnecessary movement once physically correct
DRC hole rule	0.30 mm min drill	0.20 mm min drill	Accommodate footprint hole geometry; fab capability to be verified
User silkscreen	Minimal references	Added field labels	Make the board usable without schematic lookup

13. Final Engineering Status

Stage	Status
Requirements	Complete
Architecture	Complete
Component selection	Complete with recorded footprint changes
Schematic	Complete for working PCB revision
Footprints	Assigned
Placement	Complete
Routing	Complete
Ground plane	Complete
DRC / unconnected	0 / 0
Gerber review	Complete
User silkscreen	Complete

Before fabrication. Verify final manufacturer stackup, copper thickness, controlled-drill capability, relay/terminal current ratings, thermal margin for the 5A external load target, enclosure mechanics, and BOM/MPN consistency.

14. Portfolio Documentation Notes

- Recommended GitHub repository folders: /docs, /kicad, /gerbers, /images, /bom.
- Put this engineering record in /docs as the project history and decision log.
- Use the companion datasheet/user manual as the external-facing document for someone who wants to understand or operate the board.
- Keep the final KiCad project, schematic, PCB, fabrication output, and BOM under revision control together.